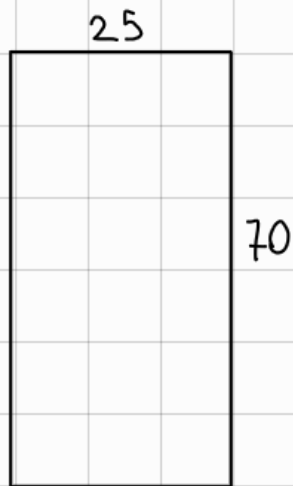
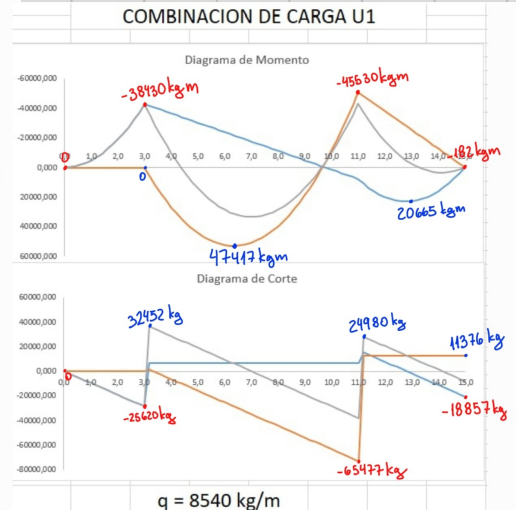
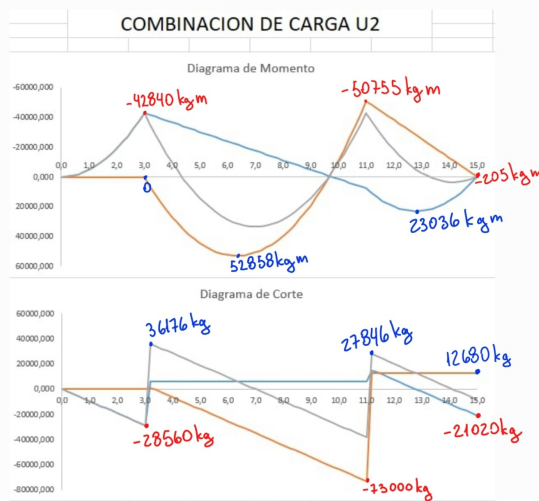
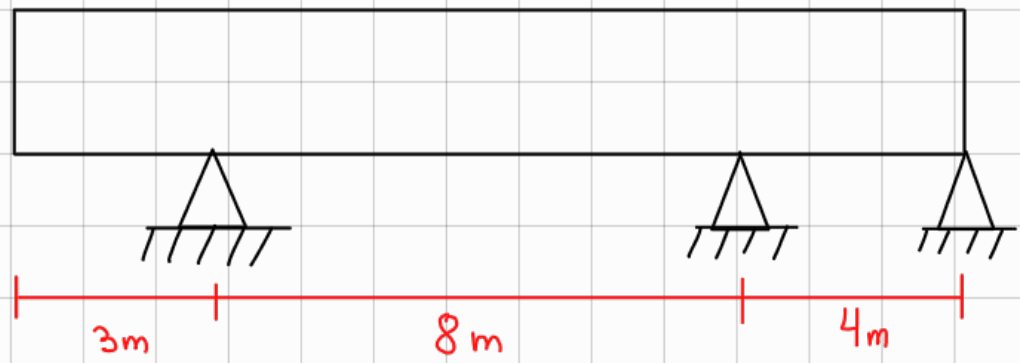


G - 20
A440-220H



$b = 25 \text{ cm}$
 $r = 40 \text{ mm}$
 $f_y = 2800 \text{ kg/cm}^2$
 $f'_c = 200 \text{ kg/cm}^2$
 $h = 70 \text{ cm}$
 $d = 66 \text{ cm}$



Armadura mínima

$$\rho_{\min} = \max \left(0,25 \sqrt{\frac{f'_c}{f_y}} ; \frac{1,4}{f_y} \right) \Rightarrow \max \left(0,25 \sqrt{\frac{20}{280}} , \frac{1,4}{280} \right) = (0,004 \text{ } \boxed{0,005}^{\text{max}})$$

$$\rho_{\min} = 0,005 \quad ; \quad A_{s_{\min}} = \rho_{\min} \cdot b \cdot d = 5 \times 10^{-3} \cdot 25 \cdot 66 = 8,25 \text{ cm}^2$$

• 4 $\phi 18$: $A_s = 10,18 \text{ cm}^2$ (No usar menos que eso)

Armadura máxima

$$\rho_{\max} = 0,75 \rho_b \quad ; \quad \rho_b = \frac{0,72 \cdot f'_c}{f_y} \cdot \frac{0,003 E_s}{f_y + 0,003 E_s}$$

$$\rho_b = \frac{0,72 \cdot 200}{2800} \cdot \frac{0,003 \cdot 2,1 \times 10^6}{2800 + 0,003 \cdot 2,1 \times 10^6} = 0,0356$$

$$\rho_{\max} = 0,026 \quad ; \quad A_{s_{\max}} = \rho_{\max} \cdot b \cdot d = 42,9 \text{ cm}^2$$

$$8,25 \text{ cm}^2 \leq A_s \leq 42,9 \text{ cm}^2$$

- Suponiendo que se usa $4\phi 18$

$$M_n = A_s \cdot f_y \left(d - \frac{1}{2} \frac{A_s \cdot f_y}{0.85 f'_c \cdot b} \right)$$

$$M_n = 10,18 \cdot 2800 \left(66 - \frac{1}{2} \frac{10,18 \cdot 2800}{0,85 \cdot 200 \cdot 25} \right)$$

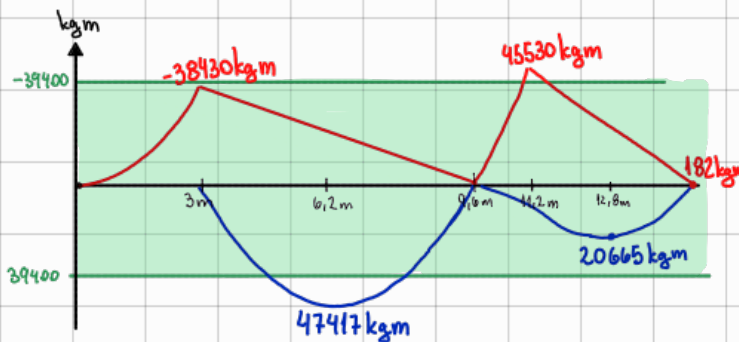
$$M_n = 1785678,35 \text{ kg}\cdot\text{cm} = 17856,8 \text{ kg}\cdot\text{m}$$

$$\phi M_n = 0,9 \cdot M_n = 16.071,1 \text{ kg}\cdot\text{m}$$

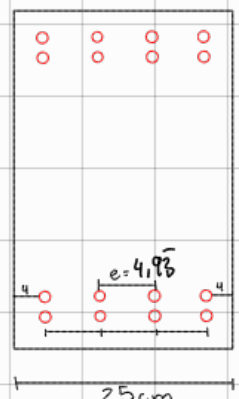
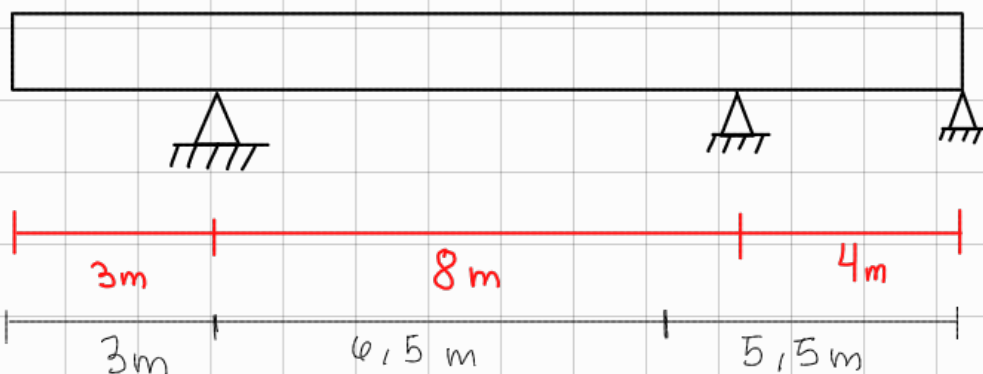
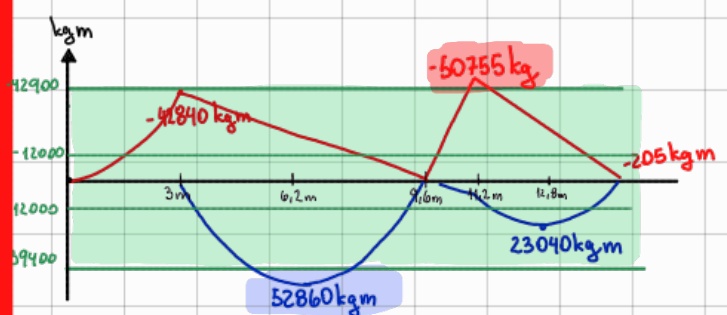
- No fue suficiente $\therefore A_s$ se aumentará a $8\phi 22$

$$\phi M_n = 42901,74 \text{ kg}\cdot\text{m}$$

Configuración U1



Configuración U2



$$* 8 \phi 22 = 30,41 \text{ cm}^2 = A_s$$

$$C_b = \min(r; e/2)$$

$$C_b = 2,46$$

$$l_d = \left(\frac{1}{1,1} \cdot \frac{f_y}{\sqrt{f'_c}} \cdot \frac{\psi_e \cdot \psi_s}{\frac{C_b + k_{tr}}{\phi_b}} \right) \phi_b = \left(1,1^{-1} \cdot \frac{280}{\sqrt{20}} \cdot \frac{\psi_e \cdot 1 \cdot 1}{\frac{246 + 0}{22}} \right) \cdot 22$$

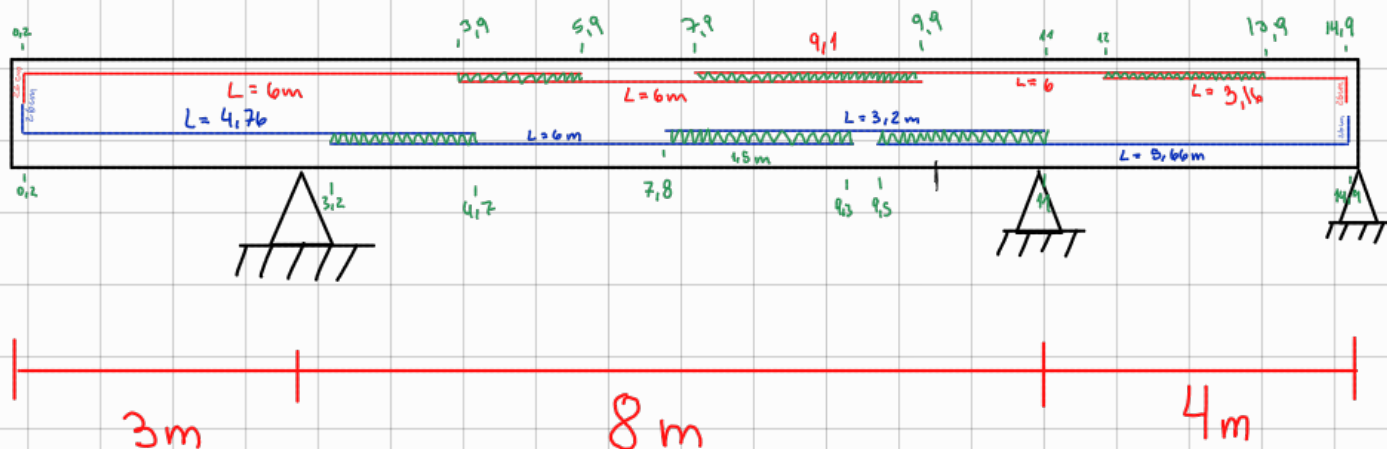
hormigon normal = 1
epoxico = 1
epoxico > 181: $\psi_s = 1$; Si no $\psi_s = 0,8$

Superior: $l_d = 1119,8 \text{ mm} \cdot \frac{\psi}{\epsilon} = 1455 \text{ mm} \approx \times 1,3 \approx 190 \text{ cm}$

Inferior: $l_d = 1119,8 \text{ mm} \cdot \frac{\psi}{\epsilon} = 1120 \text{ mm} \approx \times 1,3 \approx 150 \text{ cm}$

empalmes

Se separan para evitar
por las construcciones



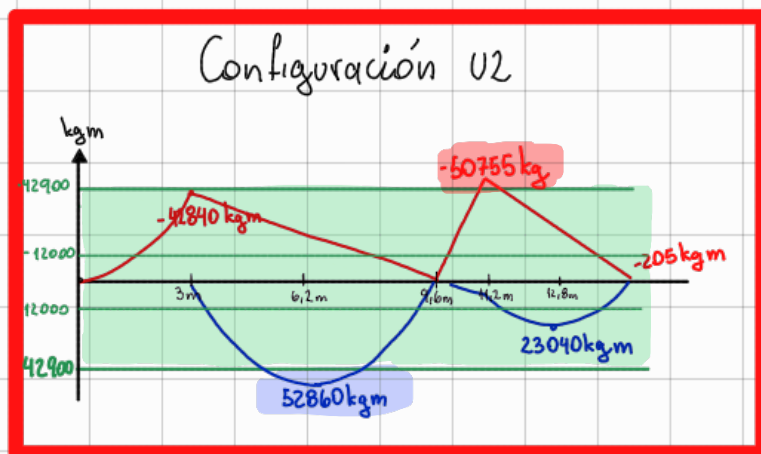
• Refuerzo

$$M_u = 53000 \text{ kgm}$$

$$= 5,3 \times 10^6 \text{ kgcm}$$

$$\phi A_s f_y \left(d - \frac{1}{2} \left(\frac{A_s f_y}{0,85 f'_c b} \right) \right) = 5,3 \times 10^6 \text{ kgcm}$$

$$A_{s_u} = 35 \text{ cm}^2$$



◦ La diferencia entre A_{s_u} y $A_s = 35 \text{ cm}^2 - 30,41 \text{ cm}^2 = 4,59 \text{ cm}^2$

$$A_{S_{\text{Refuerzo}}} = 4,59 \text{ cm}^2 \approx 2 \phi 18 = 5,09 \text{ cm}^2$$

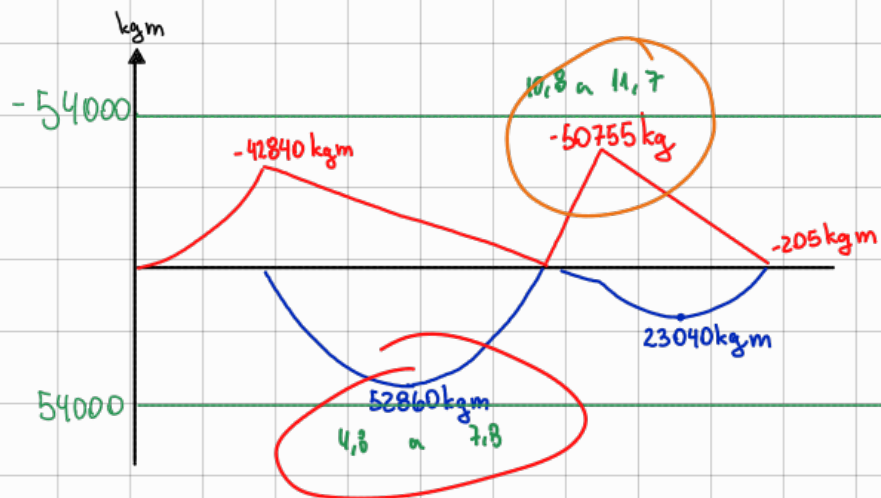


○ : $\phi 18 \text{ mm}$
 ○ : $\phi 22 \text{ mm}$

Nota: Nuevo A_s

$$\begin{aligned} A_s &= 2 \phi 18 + 8 \phi 22 \\ &= 5,09 \text{ cm}^2 + 30,41 \\ &= 35,5 \text{ cm}^2 < P_{\text{max}} = 42,9 \text{ cm}^2 \end{aligned}$$

$$\phi M_n = 53980 \text{ kg m}$$



• Agregando el refuerzo donde se excedían las limitaciones:

$2 \phi 18$ en intervalos $[4,8 \text{ a } 7,8]$ y $[10,8 \text{ a } 11,7]$ más una distancia extra para evitar que falle por adherencia

Se considerara una longitud de a lo menos 35 cm y que el largo de la barra sobrepase la zona de refuerzo 12 veces el diametro de esta.

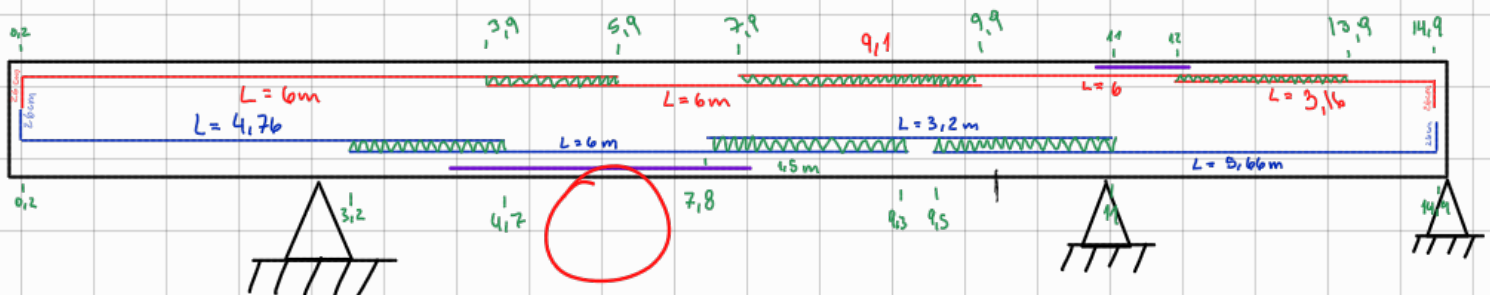


Diagrama de Corte

• Estribos

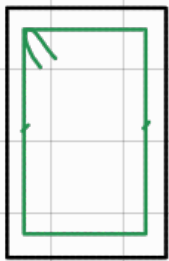
$$\phi V_n = \phi (V_c + V_s)$$



* Para la máxima licitación $V_{max} = 73000 \text{ kg}$

Se utilizará A630-420H debido que con el A440-220H no es posible satisfacer la solicitud

E $\phi 10 @ 5 \text{ cm}$ $V_c = 0,17 \lambda \sqrt{f'_c} \cdot b \cdot d = 0,17 \sqrt{20} \cdot 250 \cdot 660 = 125445 \text{ N} \rightarrow 12800 \text{ kg}$



$d = 66$
 $d/2 = 33$
 $s \leq d/2$

$$V_s = \frac{A_v \cdot f_y \cdot d}{s} = \frac{1,57 \cdot 4200 \cdot 66}{5} = 87040 \text{ kg}$$

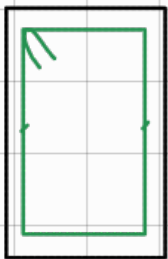
$$\phi V_n = 99840 \text{ kg} \cdot 0,75 = 74880 \text{ kg}$$

Se utilizarán en el intervalo [7m a 11,2m]

→ En 4,2m se colocarán 84 E $\phi 10 @ 5 \text{ cm}$

* Para resto $V_r = 36176 \text{ kg}$

E $\phi 8 @ 5 \text{ cm}$



$d = 66$
 $d/2 = 33$
 $s \leq d/2$

$$V_c = 0,17 \lambda \sqrt{f'_c} \cdot b \cdot d = 0,17 \sqrt{20} \cdot 250 \cdot 660 = 125445 \text{ N} \rightarrow 12800 \text{ kg}$$

$$V_s = \frac{A_v \cdot f_y \cdot d}{s} = \frac{1,01 \cdot 2800 \cdot 66}{5} = 37329 \text{ kg}$$

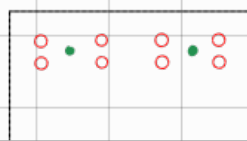
$$\phi V_n = 50129 \text{ kg} \cdot 0,75 = 37597 \text{ kg}$$

Resumen

○ : $\phi 18 \text{ mm}$
○ : $\phi 22 \text{ mm}$

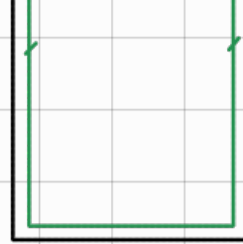
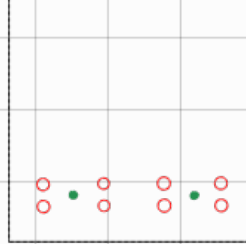
E $\phi 8 @ 5 \text{ cm}$
E $\phi 10 @ 5 \text{ cm}$

$$\phi M_n = 54000 \text{ kgm}$$



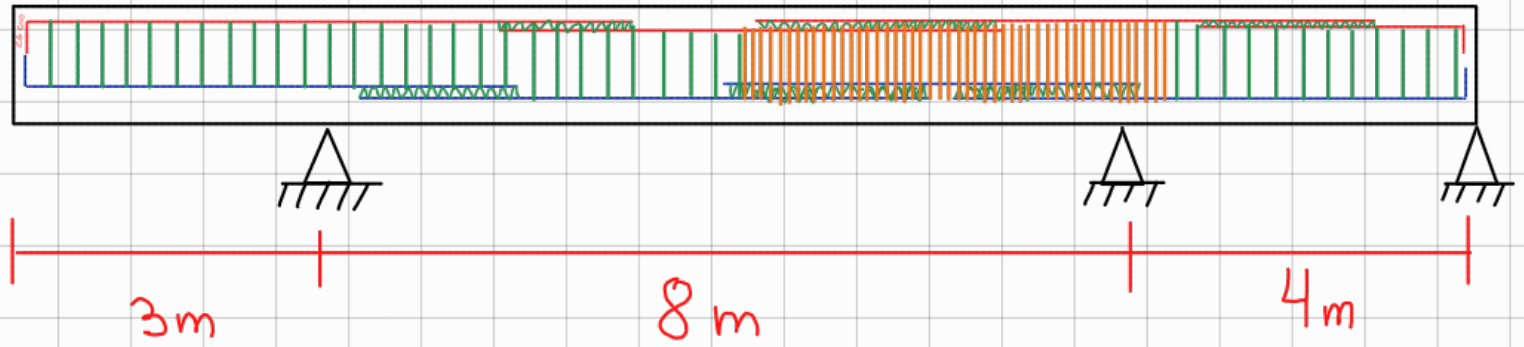
$d = 62$

$$\emptyset V_n = 101860,75 \text{ kg}$$



$$\frac{d}{2} = 31$$

$$S \leq \frac{d}{2}$$



ϕ [mm]	Área de acero (cm ²) según número de barras											
	1	2	3	4	5	6	7	8	9	10	11	12
6	0.28	0.57	0.85	1.13	1.42	1.70	1.98	2.26	2.55	2.83	3.11	3.40
8	0.50	1.01	1.51	2.01	2.52	3.02	3.52	4.02	4.53	5.03	5.53	6.04
10	0.79	1.57	2.36	3.14	3.93	4.71	5.50	6.28	7.07	7.85	8.64	9.42
12	1.13	2.26	3.39	4.52	5.66	6.79	7.92	9.05	10.18	11.31	12.44	13.57
16	2.01	4.02	6.03	8.04	10.06	12.07	14.08	16.09	18.10	20.11	22.12	24.13
18	2.55	5.09	7.64	10.18	12.73	15.27	17.82	20.36	22.91	25.45	28.00	30.54
22	3.80	7.60	11.40	15.20	19.01	22.81	26.61	30.41	34.21	38.01	41.81	45.61
25	4.91	9.82	14.73	19.64	24.55	29.45	34.36	39.27	44.18	49.09	54.00	58.91
28	6.16	12.32	18.47	24.63	30.79	36.95	43.11	49.26	55.42	61.58	67.74	73.90
32	8.04	16.09	24.13	32.17	40.22	48.26	56.30	64.34	72.39	80.43	88.47	96.52
36	10.18	20.36	30.54	40.72	50.90	61.07	71.25	81.43	91.61	101.79	111.97	122.15